

1] What is a trigger in SQL? Describe its types and when they are used.

Ans:

A trigger in SQL is a special type of stored procedure that automatically executes or fires when certain events occur in a database. Triggers are used to enforce business rules, maintain data integrity, or automate system tasks without requiring user intervention.

Triggers are associated with a specific table and are activated by certain data modification operations such as INSERT, UPDATE, or DELETE.

Types of Triggers in SQL:

1. BEFORE Trigger:

Execution Timing: It is executed before an insert, update, or delete operation is performed on the table.

Use Case: BEFORE triggers are typically used to validate or modify data before it is actually committed to the table.

Example: You can use a BEFORE trigger to ensure that a value being inserted into a table meets specific conditions.

2. AFTER Trigger:

Execution Timing: It is executed after the insert, update, or delete operation has been performed on the table.

Use Case: AFTER triggers are often used to perform tasks that should occur after the data modification, like updating a log table or cascading changes to related tables.

Example: You can use an AFTER trigger to automatically update a summary table after a new record is inserted.

3. INSTEAD OF Trigger:

Execution Timing: It is executed in place of the operation (INSERT, UPDATE, DELETE) that triggered it.

Use Case: INSTEAD OF triggers are typically used in views, allowing you to specify custom behavior for insertions, updates, or deletions that would normally not be allowed.

Example: In a view, you can define an INSTEAD OF trigger to handle how changes are propagated to the underlying tables.

Trigger Events (Actions):

Triggers can be triggered by specific events such as:

- 1. INSERT: Fired when a new row is added to a table.**
- 2. UPDATE: Fired when an existing row is updated.**
- 3. DELETE: Fired when a row is deleted from a table.**

When to Use Triggers:

Data Validation: To ensure that only valid data is inserted or updated in the database.

Audit and Logging: To automatically log changes made to the data (e.g., tracking who made the changes and when).

Cascade Updates: To automatically propagate changes in one table to related tables, maintaining data consistency.

Complex Business Logic: To enforce complex business rules that cannot be easily implemented using standard SQL constraints.